



## Safety Data Sheet

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|-------------------------------------------------|--------------------|------------------------|
| <b>Product name :</b>                           | <b>DIEKAN 1640</b> | Page : 1/6             |
| SDS n° :58156-39                                | Version :1.00      | Version of :2005-09-13 |
| This sheet supersedes the one dated :2003-06-04 |                    |                        |

### PRODUCT LABELS

|                             |                 |
|-----------------------------|-----------------|
| LABELLING (standard or EU): | Not concerned   |
| R-phrases :                 | None            |
| S-phrases :                 | None            |
| TRANSPORT LABELLING:        | Not applicable. |

### 1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY UNDERTAKING

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Name of the product :               | DIEKAN 1640                                                                                                     |
| Code No. :                          | Z22                                                                                                             |
| Product application :               | Insulating oil                                                                                                  |
| Supplier :                          | TOTAL ITALIA S.p.A.<br>Via Arconati 1<br>20135 Milano<br>ITALIA<br>Tel: +39 02 54068.1<br>Fax: +39 02 54068.291 |
| Emergency telephones :              | ORFILA / Tel : 01.45.42.59.59                                                                                   |
| See local details at end of sheet : |                                                                                                                 |
| Manufacturer :                      | TOTAL ITALIA S.p.A.<br>Via Arconati 1<br>20135 Milano<br>ITALIA<br>Tel: +39 02 54068.1<br>Fax: +39 02 54068.291 |

### 2. COMPOSITION/INFORMATION ON INGREDIENTS

#### PREPARATION

|                        |                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| Chemical nature :      | Petroleum-derived severely refined mineral-base product<br>Content in PAH according to IP 346 method < 3% |
| Composition comments : | Substances contributing to hazards :<br>None to our knowledge                                             |

### 3. HAZARDS IDENTIFICATION

|                            |                                                                             |
|----------------------------|-----------------------------------------------------------------------------|
| Health effects :           | Under normal conditions of use, the product holds no danger of intoxication |
| Environmental impact :     | Do not discharge this product into the environment.                         |
| Physico-chemical hazards : | No specific risk of fire or explosion under normal conditions of use        |

### 4. FIRST AID MEASURES

IN CASE OF SERIOUS OR PERSISTENT CONDITIONS, CALL A DOCTOR OR EMERGENCY MEDICAL CARE.

|              |                                                                                                                                                                             |
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| Inhalation : | Inhalation of heavy concentrations of vapour, fumes or spray, may cause mild irritation of the throat.<br>Transport the person into fresh air, keep warm and allow to rest. |
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|                |                                                                                                                                                         |
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| Ingestion :    | Possible risk of vomiting and diarrhoea.<br>Do not induce vomiting to avoid the risk of aspiration into the respiratory tract.<br>Give nothing to drink |
| Skin contact : | Immediately remove all soiled or stained clothing.<br>Wash the affected area immediately and repeatedly with soap and water.                            |
| Eye contact :  | Keep eyes open and rinse immediately and repeatedly with water for at least 15 minutes.                                                                 |
| Aspiration :   | If the product is believed to have entered the lungs (in case of vomiting, for example), take the person to hospital for immediate care.                |

### 5. FIRE FIGHTING MEASURES

Flash point:  
See heading 9

|                       |                                                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Extinguishing media : | - suitable:<br>Foam, carbon dioxide (CO <sub>2</sub> ), powder.<br>- not recommended:<br>do not use water jets (stick jets) for extinguishing fire since they could help to spread the flames. |
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|                    |                                                                                                                                                                                                                                                                                                                                  |
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| Specific hazards : | Incomplete combustion and thermolysis produce gases of varying toxicity such as CO, CO <sub>2</sub> , various hydrocarbons, aldehydes and soot. These may be highly dangerous if inhaled.<br>Vapours can build explosive mixtures with air.<br>Vapours are heavier than air and may spread on the ground to sources of ignition. |
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|                                        |                                                                                                               |
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| Protective measures for firefighters : | Insulated breathing apparatus must be worn in confined premises with heavy concentrations of fumes and gases. |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------|

|         |                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------|
| Other : | All combustion residues and contaminated water from fire-fighting should be disposed of according to local regulations. |
|---------|-------------------------------------------------------------------------------------------------------------------------|

### 6. ACCIDENTAL RELEASE MEASURES

See sections 8 and 13.

|                       |                                                                       |
|-----------------------|-----------------------------------------------------------------------|
| Personal protection : | Ensure good ventilation.<br>Remove sources of ignition. Do not smoke. |
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|                            |                                                                                                                                                                                                                                                                                                                                                                                 |
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| After spillage / leakage : | - On the soil:<br>Surfaces on which the product has been spilled may become slippery.<br>Do not allow the product to enter sewers or rivers or contaminate the soil.<br>Recover with mechanical means such as pumps and skimmers.<br>- On water:<br>If the product is spilled in a river or in the sewers, notify the authorities of the possible presence of surface effluent. |
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|                         |                                                                                          |
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| Spill cleanup methods : | Contain and collect the spilled product with sand or any other inert absorbent material. |
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### 7. HANDLING AND STORAGE

HANDLING :

|                               |                                                                                                                                                                                                                                                                                      |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prevention of user exposure : | Ventilate extensively if the formation of vapours, fumes, mists or aerosol is a risk.<br>Make all the necessary arrangements in order to reduce exposure risk, notably to products in use or to wastes.<br>Keep away from combustible substances; keep away from food and beverages. |
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|                                    |                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prevention of fire and explosion : | Empty containers may contain flammable or explosive vapours.<br>There is a fire hazard associated with rags, paper or any other material used to remove spills which become soaked with product.<br>Avoid accumulation of these: they are to be disposed off safely after use. |
| Precautions :                      | Avoid static electricity build up with connection to earth.<br>Set up machinery and equipment so as to avoid the risk of accidental spills or splashes onto hot machine parts and electrical contacts (on joint failure, for example).                                         |
| <b>STORAGE :</b>                   |                                                                                                                                                                                                                                                                                |
| Technical measures :               | Make the necessary arrangements to prevent water and soil pollution.                                                                                                                                                                                                           |
| Storage precautions :              | - Suitable:<br>Store at room temperature, protected against contact with water and moisture, and away from any source of ignition.<br>- To be avoided:<br>Do not store exposed to the elements.                                                                                |
| Incompatible products :            | Dangerous reaction with strong oxidizing agents.                                                                                                                                                                                                                               |
| Packaging materials :              | - Recommended:<br>Use only hydrocarbon-resistant containers, joints, pipes, etc.<br>Keep in original container if possible.                                                                                                                                                    |

### 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technical measures :                              | Use the product in a properly ventilated atmosphere.<br>When working on enclosed place (tanks, reservoirs...), make sure that atmosphere is not suffocating and/or wear recommended equipment.                                                                                                                                                                                                                                                                                                                                                                                                               |
| Occupational exposure limit :                     | . oil mist : 10mg/m3, for 15 minutes<br>. oil mist : 5mg/m3, for 8 hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Hand protection :                                 | Impermeable hydrocarbon-proof gloves.<br>recommended material: nitrile , neoprene.<br>The break through times of the same type of glove of different manufacturers can be very different - even if the layer thickness is similar. Therefore the break through times have to be found out from the manufacturer of the protective gloves themselves.<br>The demands on the gloves are determined by the conditions in practice (e.g. multiple use, mechanical load, temperature, strength and duration of exposition).<br>Before choosing suitable gloves, it is recommended that the user tests the gloves. |
| Eye protection :                                  | Goggles, in case of risk of splashing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Skin and body (other than the hands) protection : | As required, wear a face mask, hydrocarbon-proof clothing, and safety boots (when handling drums).<br>Don't wear rings, watches or anything similar which can retain the product and may give rise to skin conditions.                                                                                                                                                                                                                                                                                                                                                                                       |
| Hygienic work practices :                         | Avoid prolonged and repeated contact with the skin, especially with used or waste product<br>If the product comes into contact with the skin, wash the affected area immediately and copiously with soap and water.<br>Do not use abrasives, solvents or fuels.<br>Do not put the product-soaked rags into the pockets of working clothes.<br>Do not eat, drink or smoke while handling the product.<br>Do not use cloths stained with the product to dry hands.<br>Immediately remove all soiled or stained clothing.                                                                                       |

### 9. PHYSICAL AND CHEMICAL PROPERTIES

|              |        |
|--------------|--------|
| Appearance : | Liquid |
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|                            |                                                                                                          |
|----------------------------|----------------------------------------------------------------------------------------------------------|
| Colour :                   | Light yellow.                                                                                            |
| Odour :                    | Characteristic.                                                                                          |
| Density/specific gravity : | 853 kg/m3<br>Temperature (°C) 20                                                                         |
| Flash point :              | 174 ° C (ASTM D 93)                                                                                      |
| Solubility :               | - in water :<br>Insoluble and immiscible.<br>- in organic solvents :<br>Soluble in many common solvents. |
| Viscosity :                | c.ca 13 mm2/s<br>Temperature (°C) 40                                                                     |

### 10. STABILITY AND REACTIVITY

|                              |                                                                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Stability :                  | The product is stable at normal storage, handling and use temperatures.                                                           |
| Conditions to avoid :        | Heat (temperatures above flash point), sparks, ignition points, flames, static electricity                                        |
| Materials to avoid :         | Avoid contact with strong oxidizers                                                                                               |
| Hazardous decomp. products : | Incomplete combustion and thermolysis produces potentially toxic gases such as CO, CO2, various hydrocarbons, aldehydes and soot. |

### 11. TOXICOLOGICAL INFORMATION

|                                          |                                                                                                                                                                   |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute toxicity / Local effect :          |                                                                                                                                                                   |
| Inhalation, comments:                    | Risk is improbable under normal conditions of use<br>Inhalation of high concentrations of vapour or aerosols may cause irritation of the upper respiratory tract. |
| Skin contact, comments:                  | Risk is improbable under normal conditions of use.                                                                                                                |
| Ingestion, comments:                     | In case of ingestion of small quantities, no important effect observed. in case of ingestion of larger amounts: abdominal pain, diarrhea, ...                     |
| CHRONIC TOXICITY OR LONG-TERM TOXICITY : |                                                                                                                                                                   |
| Skin contact :                           | Characteristic skin affections (oil blisters) may develop following prolonged and repeated exposure through contact with stained clothing                         |

### 12. ECOLOGICAL INFORMATION

|                                 |                                                                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Comments about ecotoxicity :    | Experimental data on the finished product are not available.<br>It is considered to present a little danger for aquatic life.<br>no information available for used product                                                                               |
| Mobility :                      | - Air:<br>there is a slow loss by evaporation.<br>- Soil:<br>Given its physical and chemical characteristics, the product generally shows little mobility in the ground.<br>- Water:<br>The product is insoluble; it spreads on the surface of the water |
| Persistence and degradability : | No experimental information about the finished product.<br>The product should offer slow biodegradability                                                                                                                                                |



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### 13. DISPOSAL CONSIDERATIONS

Waste disposal :

Dispose of in a safe manner, in accordance with local regulations.  
If need be, collection by an authorised waste contractor and regeneration or incineration at an approved installation.

Disposal of contaminated packaging :

Proceed in compliance with the prevailing regulations.

National regulations :

See Legislative Decrees February 5, 1997 n. 22, DPR August 23, 1982 n. 691 their updates and eventual regional provisions for the disposal of dangerous special waste.

### 14. TRANSPORT INFORMATION

Not concerned by transport regulations RID/ADR, ADNR, IATA, IMDG.

Road (ADR) / Rail (RID) :

Transport by barge (ADNR) :

Marine (IMO-IMDG) :

Air (ICAO/IATA) :

### 15. REGULATORY INFORMATION

The product has not to be labelled according to EC-directions / hazardous materials ordinance.

In compliance with Legislative Decree of March 14, 2003 n. 65.

Symbol : none

Risk phrases :

None

Safety phrases :

None

Other :

Safety data sheet is available on request for professional users

EU directives :

Hazardous preparations directive 1999/45/EC modified (Directive 2001/60/EC).  
This product fulfills the requirements of the European directives :  
76/769/EEC  
2000/53/EC  
2002/95/EC  
2002/96/EC  
2003/11/EC

National regulations (Italy) :

DPR n.303/1956 General regulations for job health.  
Legislative Decree of September 19, 1994, N°626 concerning improvement of worker health and safety on working place.  
DPR 336/1994 Regulation on new lists of industry and agriculture professional illness.  
D.Lgs 242/96 " Change and integration of D.Lgs 626/94 about European Directions concerning improvement of health and safety for the workers while are working. "  
D.Lgs 25/02 " Concerning the improvement of health and safety for the workers while are working. "  
Decree June 14, 2002 - Implementation of the Directive 2001/59/EC, relating to the 28th edition at the technical progress of the Directive 67/548/EC concerning dangerous preparations classification and labelling.  
Decree September 7, 2002 - Implementation of the Directive 2001/58/CE, concerning terms of information about dangerous substances and preparations on the market.  
Legislative Decree March 14, 2003 n.65 - Putting into practice of the Directives 1999/45/EC and 2001/60/EC concerning dangerous preparations classification, packing and labelling.



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### 16. OTHER INFORMATION

This sheet is in compliance with the standards defined by the directives 91/155/CEE, 93/112/CEE, 2001/58/CE and the article 14 of the directive 1999/45/EC.

Revision date: 2005-09-13

Supersedes the data sheet of: 2003-06-04

\* Information revised since the previous version of the SDS :

SDS No. : 260.04

Emergency number : For France, in case of poisoning call the Antipoison Centre (if possible in your area) and/or the SAMU (15), see ORFILA in heading 1 - Tel : Angers 02.41.48.21.21 - Bordeaux 05.56.96.40.80 - Lille 0 825 812 822- Lyon 04.72.11.69.11 - Marseille 04.91.75.25.25 - Nancy 03.83.32.36.36 - Paris 01.40.05.48.48 - Rennes 02.99.59.22.22 - Strasbourg 03.88.37.37.37 - Toulouse 05.61.77.74.47.

This safety data sheet serves to complete but not to replace the technical product sheets. The information contained herein is given in good faith and is accurate to the best of knowledge at the date indicated above. It is understood by the user that any use of the product for purposes other than those for which it was designed entails potential risk. The information given herein in no way dispenses the user from knowing and applying all provisions regulating his activity. The user bears sole liability for the precautions required when using the product. The regulatory texts indicated herein are intended to aid the user to fulfil his obligations. This list is not to be considered complete and exhaustive. It is the user's responsibility to ensure that he is subject to no other obligations than those mentioned.